

Deterministic six-window discrepancy for short permutation families

Research proof candidate; independent review pending

30 September 2026

For a probability distribution μ supported on q permutations of $[n]$, define, on distinct labels,

$$\delta(a, b, c) = \frac{1}{2} - \frac{3}{2} \Pr_{\rho \sim \mu}(\rho(b) < \rho(a), \rho(b) < \rho(c)).$$

Thus $|\delta| \leq 1$, $\delta(a, b, c) = \delta(c, b, a)$, and the sum over the three choices of center is zero. For a target permutation σ , put

$$F(\sigma) = \sum_{j=1}^{n-2} \delta(\sigma_j, \sigma_{j+1}, \sigma_{j+2}).$$

This is a constant-position-weight version of the triple contribution in the palindrome construction. The conclusions below are about this linear statistic, not the full nonlinear fair-order probabilities.

Theorem 1. *There are absolute constants $C, c > 0$ such that, if $n \geq Cq$, some six distinct labels have an interior middle-pair swap which changes their four-triple path score by at least $c/\sqrt{q}$. If $n \geq 2Cq$, then*

$$\max_{\sigma} F(\sigma) - \min_{\sigma} F(\sigma) \geq c \frac{n}{\sqrt{q}}.$$

The second conclusion can be witnessed on a family of independent swaps in disjoint six-label windows.

For any six labels define the swap difference

$$\begin{aligned} D(a, b, c, d, e, f) = & \delta(a, b, c) + \delta(b, c, d) + \delta(c, d, e) + \delta(d, e, f) \\ & - \delta(a, b, d) - \delta(b, d, c) - \delta(d, c, e) - \delta(c, e, f). \end{aligned}$$

Lemma 2 (Small swaps force a pair coboundary). *Let δ be any function on distinct triples with the symmetry, center-sum identity, and bound $|\delta| \leq 1$ stated above. Suppose $|D(a, b, c, d, e, f)| \leq \eta$ for every six distinct labels. Then there is an antisymmetric array $a_{ij} = -a_{ji}$, with $|a_{ij}| \leq 3/2$, such that, for all distinct b, c, d ,*

$$|\delta(b, c, d) - a_{cb} - a_{cd}| \leq \eta + \frac{128}{n}.$$

Proof. Extend δ to be zero when labels repeat, so all its identities remain valid. All averages in this proof are uniform over all n labels, with replacement. Define

$$U_b(c) = \mathbb{E}_a \delta(a, b, c), \quad r_b = \mathbb{E}_c U_b(c), \quad s_c = \mathbb{E}_b U_b(c).$$

Averaging the center-sum identity gives $s_c = -r_c/2$ and $\mathbb{E}_c r_c = 0$. Set

$$K_b(c, d) = \delta(b, c, d) - \delta(b, d, c) + U_b(c) - U_b(d),$$

$$V(c, d) = \mathbb{E}_b K_b(c, d) = U_c(d) - U_d(c) - r_c/2 + r_d/2.$$

For fixed distinct b, c, d, e , averaging $D(a, b, c, d, e, f)$ over a, f gives exactly $K_b(c, d) - K_e(c, d)$. The probability that the six labels are not distinct is at most $9/n$, and $|D| \leq 8$. Consequently

$$|K_b(c, d) - K_e(c, d)| \leq \eta + 72/n.$$

Averaging over e , including the at most three exceptional values, and using $|K_b - K_e| \leq 8$, yields, for distinct b, c, d ,

$$|K_b(c, d) - V(c, d)| \leq \zeta := \eta + 96/n. \quad (2)$$

For distinct b, c , average (2) over d . The exceptional two values contribute at most $14/n$, since $|K_b - V| \leq 7$. Directly averaging the definitions gives

$$|2U_b(c) + 2U_c(b) - r_b - r_c| \leq \zeta + 14/n =: \zeta_2. \quad (3)$$

Indeed, $\mathbb{E}_d [K_b(c, d) - V(c, d)]$ equals the expression inside the absolute value in (3). Put

$$S_{bc} = U_b(c) + U_c(b) - (r_b + r_c)/2, \quad a_{bc} = \frac{U_b(c) - U_c(b)}{2} + \frac{r_c - r_b}{4}.$$

Then $|S_{bc}| \leq \zeta_2/2$, a is antisymmetric, and $|a_{bc}| \leq 3/2$. Use the center-sum identity to write $3\delta(b, c, d)$ as the sum of its differences with the other two

centers. Apply (2) to each difference. After substituting the definitions, this gives

$$3(\delta(b, c, d) - a_{cb} - a_{cd}) = -\frac{1}{2}S_{cb} - \frac{1}{2}S_{cd} + S_{bd} + E, \quad |E| \leq 2\zeta.$$

The absolute value is at most $\zeta_2 + 2\zeta$, proving the stated bound (in fact $\eta + 101/n$ is enough). $\square$

Proof of Theorem ??. Let η be the largest absolute six-label swap difference, and take a from Lemma ??. Write $m = n - 1$, $\tau = \eta + 128/n$, and $p_{ij} = \Pr(\rho(i) < \rho(j))$. Define

$$d_{ij} = p_{ij} - \frac{1}{3} + \frac{4}{3}a_{ij}.$$

Antisymmetry and $p_{ij} + p_{ji} = 1$ imply $d_{ij} + d_{ji} = 1/3$. Hence some i satisfies $\sum_{j \neq i} d_{ij} \geq m/6$.

As before, form the Gram matrix G of the m indicator vectors $\mathbf{1}_{\rho(i) < \rho(j)}$ in the weighted q -dimensional space. It has rank at most q , diagonal p_{ij} , and off-diagonal

$$G_{jk} = \frac{1}{3} - \frac{2}{3}\delta(j, i, k) = \frac{1}{3} - \frac{2}{3}(a_{ij} + a_{ik}) + E_{jk}, \quad |E_{jk}| \leq \frac{2}{3}\tau.$$

Subtract the displayed rank-at-most-two additive matrix, including its diagonal, to obtain $B = \text{diag}(d_{ij}) + E$, where E has zero diagonal. Then B is symmetric with rank at most $q + 2$, $\text{tr } B \geq m/6$, and $|d_{ij}| \leq 8/3$. Thus

$$\frac{m}{6} \leq \sqrt{q+2} \|B\|_F \leq \sqrt{q+2} \left(\frac{8}{3}\sqrt{m} + \frac{2}{3}\tau m \right).$$

It follows that

$$\eta \geq \frac{1}{4\sqrt{q+2}} - \frac{4}{\sqrt{n-1}} - \frac{128}{n}.$$

For example, $n \geq 2^{16}q + 1$ makes the right side at least $1/(64\sqrt{q})$.

The argument is hereditary under restricting to unused labels. Greedily choose disjoint six-label windows while at least $2^{16}q + 1$ labels remain. In each, permit only the swap of its middle two labels; the outer two labels on each side stay fixed. Every affected triple lies entirely in that window. On the resulting binary family of target permutations, F is therefore a constant plus a sum of independent binary choices with coefficients D_j . Its range is exactly $\sum_j |D_j|$. For n at least twice a sufficiently large constant times q , there are at least cn windows, proving the result. $\square$

Next obstacle. For the actual palindrome first projection, binomial weights vary with the target position. On a fixed six-window, adjacent central weights have relative total-variation change $O(n^{-1/2})$, while the effective number of blocks is about $M/\sqrt{n}$. Truncation to a weighted interval therefore suggests the same $n^{13/10}$ power barrier for the linear projection, up to logarithms. This extension and, especially, control of the full deterministic nonlinear remainder are not asserted here.